

SOLID WASTE MANAGEMENT (22CV563)

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MODULE 2

Waste Generation and Characterization: Factors affecting waste generation and methods to estimate the quantity of waste generated. Physical, Chemical and Biological methods of waste characterization

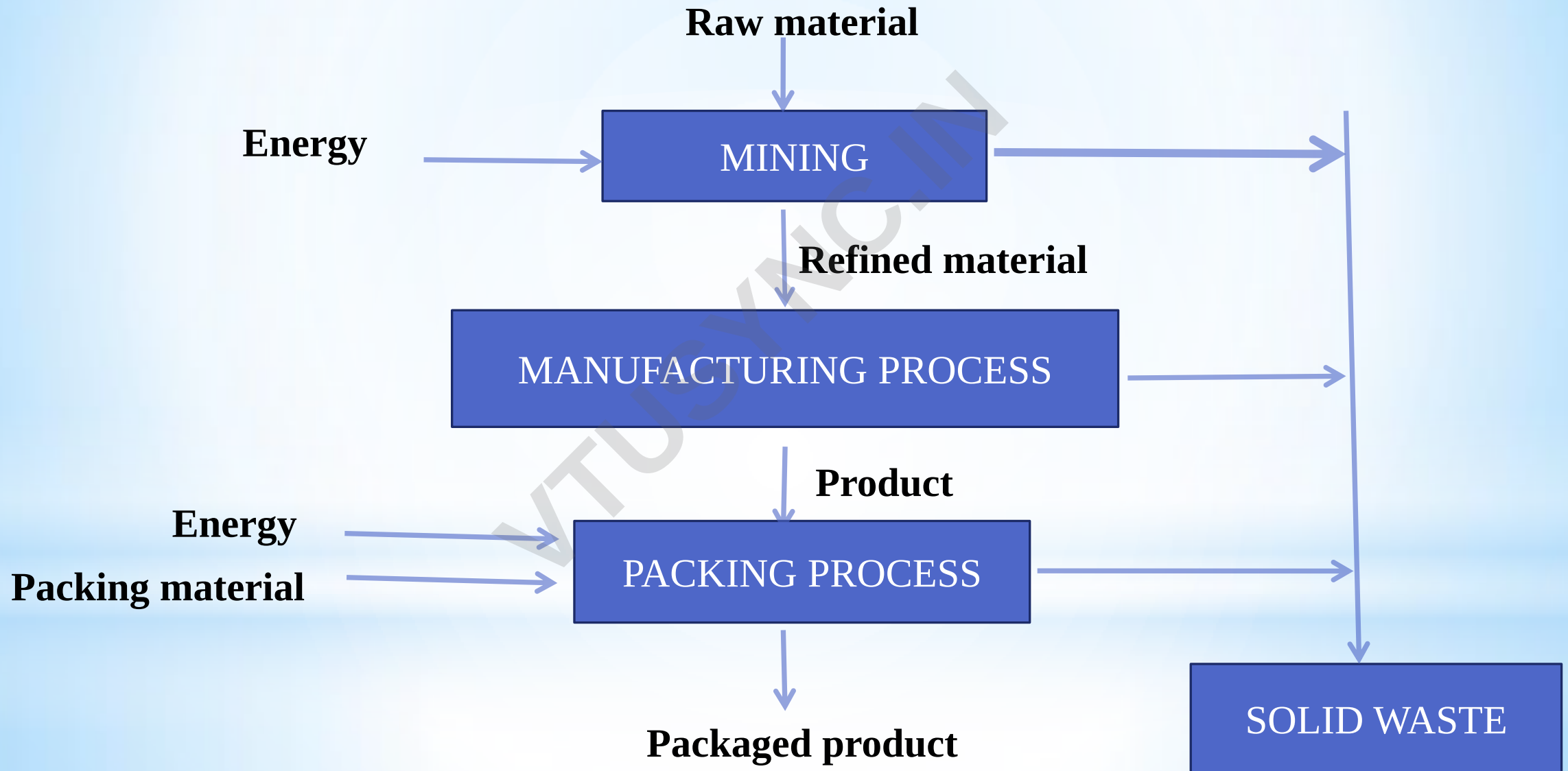
WASTE GENERATION

Every individual produces around **0.4-0.6kg** of garbage everyday. The activities that produce garbage in **URBAN SOCIETY** by,

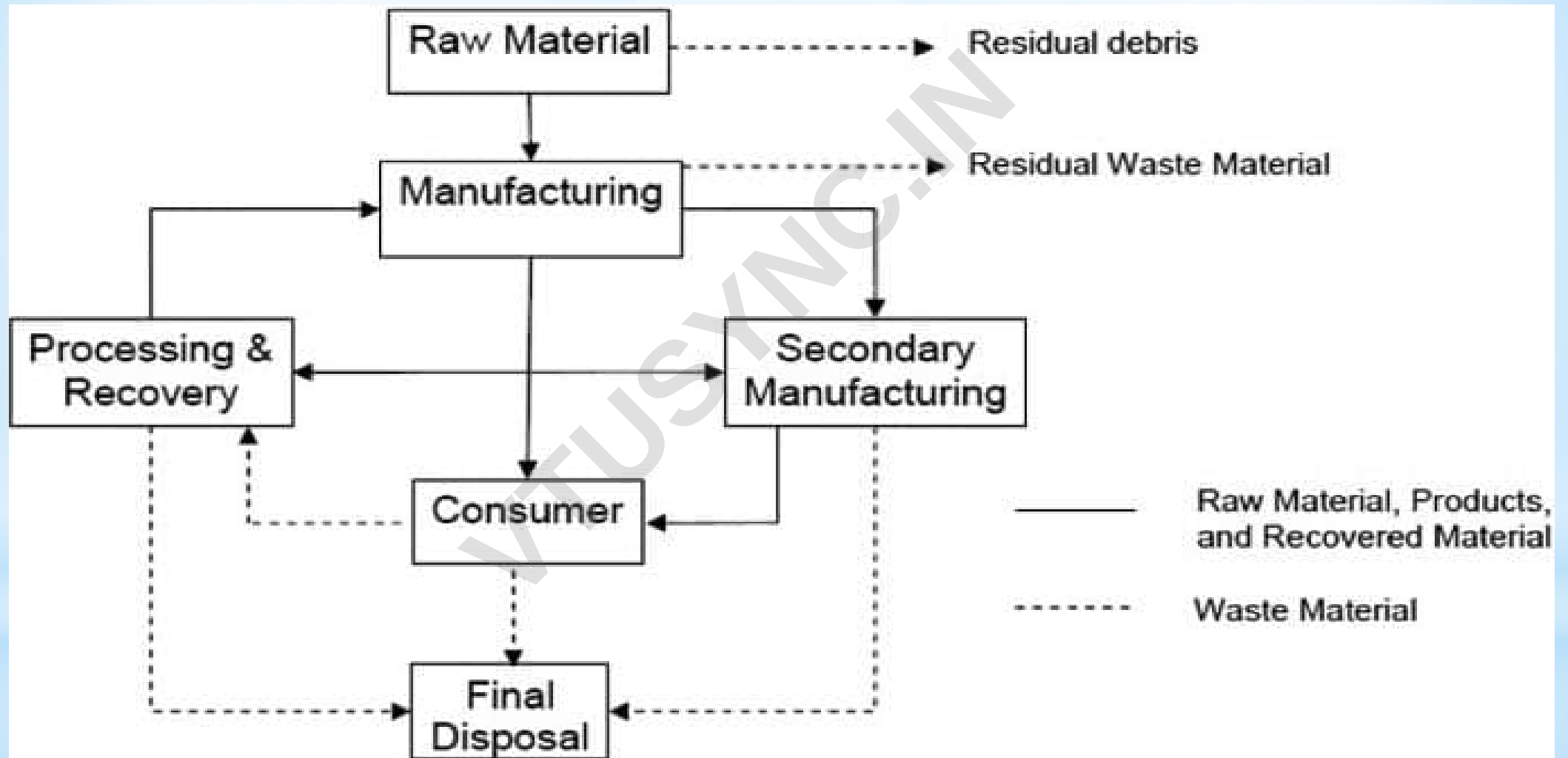
- Preparing, cooking and serving food
- Sweeping of house
- Discarding waste like paper, packings, plastics etc.
- Cleaning of lawns and gardens
- Construction and demolition activities
- Markets, commercial sources, community halls, treatment plants etc..



SW GENERATION BY TECHNOLOGICAL PROCESS



MATERIAL FLOW AND WASTE GENERATION



Importance of Waste Quantities

The waste quantities of solid waste generated and collected are of critical importance,

- To determining compliance with central and state waste diversion program
- In selecting specific equipment
- In designing of waste collection routes
- Materials Recovery Facilities (MRF)
- In designing disposal facilities



METHODS TO DETERMINE WASTE GENERATION RATE

Most commonly used methods to estimate the waste quantities are,

- 1. Load-Count Analysis**
- 2. Weight-Volume Analysis**
- 3. Materials-Balance Analysis**

1. Load-Count Analysis:

- **Approach:** This method involves **counting** the number of waste loads (trucks or containers) arriving at a disposal or processing site. Based on the **type and size of each load**, an estimate of the waste quantity is made.
- **Accuracy:** It provides a **rough estimate**,
- **Data Used:** The number of loads and their estimated vol or wt **based on load sizes**.
- **Advantages:** **Simple and quick** to implement, no specialized equipment required.
- **Disadvantages:** **Less accurate** & It relies heavily on **assumptions** about the waste.



2. Weight-Volume Analysis:

- **Approach:** This method directly measures either the **weight or volume** of waste, often using **weighbridges** to measure truckloads.
- **Accuracy:** More accurate than load-count, as it involves direct measurement of waste.
- **Data Used:** Direct measurements of waste **wt (tons)** or **vol (m³)** using weighing systems.
- **Advantages:** **High accuracy**, particularly when waste is measured by weight.
- **Disadvantages:** Requires equipment such as weighbridges or calibrated containers, making it **more resource-intensive**.



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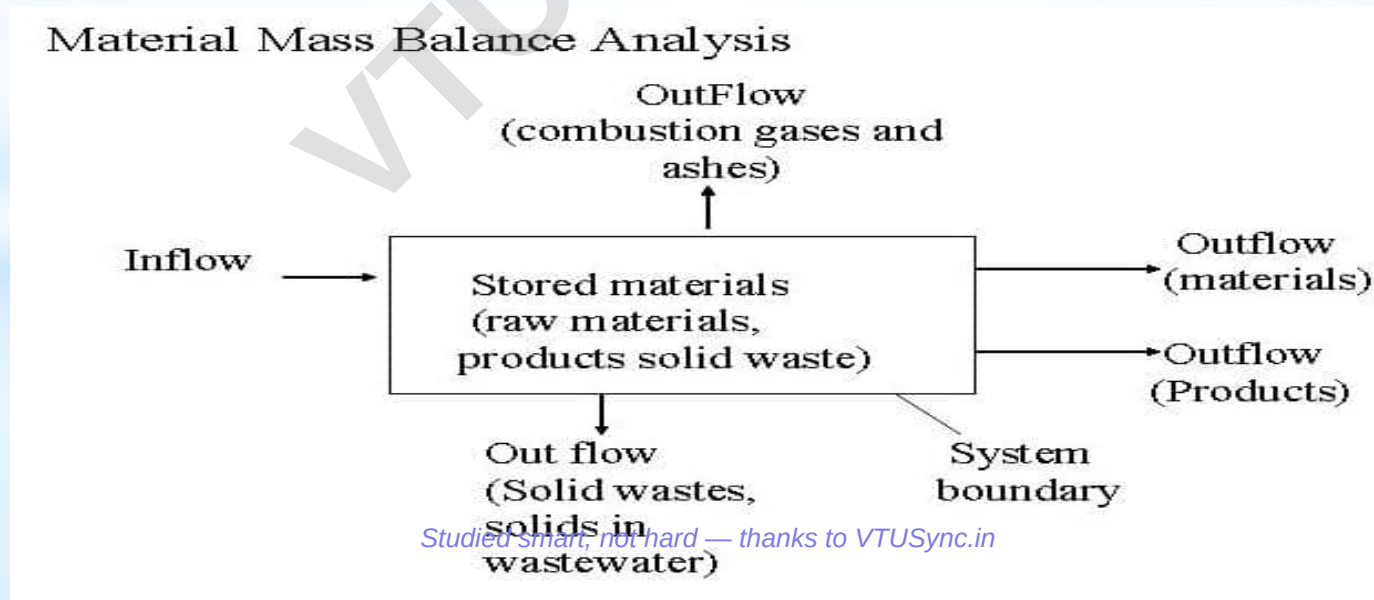


3. Material Balance Method (Material Flow Analysis - MFA):

Approach: This method estimates waste generation **by tracking** the input and output of materials within a defined system (e.g., a city, factory, or industry). It is based on the principle that the **total material entering** a system (raw materials and imports) minus the materials **leaving the system** (products, exports, emissions, and waste) gives an estimation.

Process:

- Calculate the input materials (e.g., raw materials, energy, water).
- Subtract the output in terms of products, exports, emissions, and recyclable materials.
- The difference represents the waste generated.



Accuracy: This method can provide a **comprehensive and accurate** estimate of waste generation, especially when **detailed data** on material flows are available.

Data Used: Information on material inputs (raw materials, consumables) and outputs (finished products, exports, emissions, and waste).

Advantages:

- Can account for **complex waste streams**, including byproducts, emissions, and unaccounted losses.
- Helps **track** resource use and sustainability.

Disadvantages:

- Requires **extensive and detailed data** on material flows, which may not always be available.
- **More complex and resource-intensive** to implement compared to load-count analysis

Factors affecting waste generation

➤ GEOGRAPHICAL LOCATION

Ex: Southern India and Northern India Climatic condition

➤ SEASON

Ex: Harvesting season and growing season

➤ FREQUENCY OF COLLECTION

Unlimited collection service is provides more waste collection

This doesn't mean that more waste are generated.

➤ CHARACTERISTICS OF POPULATION

Ex: Low income and High income family

➤ EXTENT OF SALVAGING AND RECYCLING

Existing of saving and recycling operation in communities

➤ PUBLIC ATTITUDE

Willing to change of their own to conserve nature

➤ LEGISLATIONS

For use and disposal of specific materials



Properties of solid waste

PHYSICAL CHARACTERISTICS

1. INDIVIDUAL COMPONENTS:

Components are paper, plastic, cardboard, textile, glass etc.,

Analyse the behavior of each components like MC, density, wt and distribution %

2. PARTICAL SIZE:

Graphical representation of analysis used in designing machineries

3. MOISTURE CONTENT:

Cost of collection, trasport and treatment depends on MC value

$$MC = ((a-b)/a) \times 100$$

where, a is intial weight of the waste

b is dry weight of the waste (@105 dergee C)

4. DENSITY:

Design of landfills, transport vehicles depends on optimum density

CHEMICAL CHARACTERISTICS

1. LIPIDS:

Includes oils, fats and grease.

Prime sources are garbage, automobiles, refineries.

High heating value = 38000 KJ/Kg

2. CARBOHYDRATES:

Sugar and polymers of sugar(cellulose, starch etc)

Highly biodegradable

3. PROTIENS:

Contains C,H, O and N

Prime sources are garbage and garden waste

4. NATURAL FIBERS:

Cellulose, natural compounds and lignins

Highly combustible

5. SYNTHETIC ORGANIC MATERIAL:

Plastics (non biodegradable)

High Calorific value and its effects

6. HEATING VALUE (ENERGY CONTENT)

Determined by Bomb Calorimeter Test

Provides the heating index of waste

Energy value as “DRY BASIS”, KJ/Kg = unit energy value $\times (100 / (100 - \% \text{moisture}))$

Energy value as “DRY BASIS”, KJ/Kg o “ASH FREE” = unit energy value $\times (100 / (100 - \% \text{Ash} - \% \text{moisture}))$

7. ULTIMATE ANALYSIS:

Analysis of C, H, O, S proportions

To make Mass Balance calculation for chemical and thermal process

$$\text{KJ/Kg} = 337 \text{ C} + 1428 (\text{H} - \text{O}/8) \times 9\text{S}$$

8. PROXIMATE ANALYSIS

- Moisture (loss of moisture when heated to 105 deg C for 1 hr)
- Volatile Combustible matter (additional loss of weight on ignition at 950 deg C)
- Fixed Carbon (residue after volatile matter is removed)
- Ash (wt of residue combustion in a open crucible)

8. FUSING POINT OF ASH:

burning of waste forms a solid (CLINDER) by fusion and agglomeration

fusing temp : 1100 to 1200 deg C

PARTICLE SIZE: Particle size of solid waste is important in recovery of materials.

Particle size distribution is done by means of screens & magnetic separators

MOISTURE CONTENT: Expressed as moisture content per unit mass of wet or dry material.

$$\text{Moisture content} = \left(\frac{a-b}{a} \right) 100$$

Where, a = Initial mass of sample delivered
 b = Mass of sample after drying

DENSITY: Density of solid waste vary with

- * Geometrical location
- * Season of the year
- * length & time of storage.

CHEMICAL PROPERTIES OF SOLID WASTES

Chemical properties or composition of SW is important in evaluating alternative process and energy recovery options

If solid wastes are to be used as fuel the imp properties to be known are:

- 1) Proximate analysis { moisture
- Ash
- 2) Fusion point of ash
- 3) Analysis of C, H, N, S
- 4) Heating value.

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ENERGY CONTENT:

Equations used for calculation of energy content on dry basis: (on discarded basis)

$$\text{KJ/kg} = \frac{100}{100 - \% \text{ moisture}}$$

ON THE BASIS OF FREE DRY BASIS .

$$\text{KJ/kg as dry basis} = \frac{100}{100 - \% \text{ ash} - \% \text{ of moisture}}$$

ON THE BASIS OF CHEMICAL CONTENT

If energy values are not available energy content can be calculated by using modified DULONG'S formula

$$\text{KJ/kg} = 337C + 1426 \left[H - \frac{O}{8} \right] + 95S$$

METHODS OF QUANTIFICATION OR SAMPLING PROCEDURES

It is difficult to predict composition of solid waste & and hence difficult to design and operate S.W.M this is due to heterogeneous nature of waste

Following methods are used to assess components of solid waste:

- 1) Unload the load of a truck away from other operations
- 2) Quarter the waste load
- 3) Quarter the one quarter & quarter that quarter
- 4) Select one quarter & separate all individual components

9

- 5) place the separated components in a container of known volume and measure volume of each component
- 6) Determine the percentage distribution of each component by mass
- 7) 100 to 200 kg of waste should be stored to obtain representative sample
- 8) To obtain accuracy samples are collected at each season of the year.
- 9) Prepare the projections

PROBLEMS

- 1) Estimate the density of waste sample on as discarded basis take 1000kg sample.

COMPONENT	% By MASS	DENSITY TYPICAL (kg/m ³)
Food waste	20	300
Paper	40	100
plastics	05	90
Garden trimming	15	150
wood	05	250
Tin cans	05	100
card board	10	80

$$\text{Density} = ?$$

$$\text{Density} = \frac{M}{\text{volume}}$$

$$\text{Density} = \frac{1000}{x}$$

COMPONENT	% BY MASS	DENSITY TYPICAL kg/m ³	VOLUME BASED ON 1000kg SAMPLE
Food waste	20	300	$\frac{0.2}{300} \times 1000 = 0.66$
Paper	40	100	$\frac{0.4}{100} \times 1000 = 0.4$
plastics	05	90	$\frac{0.05}{90} \times 1000 = 0.55$
Garden trimming	15	150	$\frac{0.15}{150} \times 1000 = 1.0$
wood	05	250	$\frac{0.05}{250} \times 1000 = 0.2$
Tin can	05	100	$\frac{0.05}{100} \times 1000 = 0.5$
card board	10	80	$\frac{0.10}{80} \times 1000 = 1.25$

$$\Sigma V = 8.16$$

$$\text{Density} = \frac{1000}{8.16} = 122.54 \text{ kg/m}^3$$

27 Estimate the density of a town having different kinds of waste considering a total volume of 1000kg. The typical values of densities and its % mass is given in the table.

COMPONENT	% BY MASS	DENSITY TYPICAL kg/m ³	VOLUME BASED
Food waste	15	290	0.517
paper	45	85	5.29
plastic	10	50	2
yard trimming	10	65	1.538
wood	10	105	0.952
Tin cans	05	240	0.208
card board	05	90	0.555

$$\Sigma V = 11.06$$

$$\text{DENSITY} = \frac{1000}{11.06} = 90.415 \text{ kg/m}^3$$

(10)

37 Estimate total density of solid waste sample having the following properties given in the table. Take sample of volume as 1000 kg

COMPONENT	% BY MASS	DENSITY TYPICAL kg/m^3	VOLUME BASED ON 1000 kg SAMPLE
Food waste	25	310	0.806
Paper	35	75	4.667
plastic	05	45	1.111
Yard trimming	05	75	0.667
Wood	20	95	2.105
Tin cans	05	220	0.227
Card boards	05	70	0.714

$$\Sigma V = 10.297$$

$$\text{DENSITY} = \frac{1000}{10.297} = 97.115 \text{ kg/m}^3$$

47 Estimate the energy content of a solid waste sample on unit energy content of 100 kg sample on dry basis and ash free dry basis (Assume 5% ash)

COMPONENT	% BY MASS	MOISTURE CONTENT	ENERGY CONTENT	DRY MASS (kg)	TOTAL ENERGY CONTENT
FOOD WASTE	15	70	4650	$\frac{100-70}{100} \times 15 = 4.5$	$4650 \times 15 = 69750$
PAPER	45	06	16750	$\frac{100-6}{100} \times 45 = 42.3$	$16750 \times 45 = 753750$
CARD BOARD	10	05	16300	$\frac{100-5}{100} \times 10 = 9.5$	$16300 \times 10 = 163000$
plastics	10	02	32600	$\frac{100-2}{100} \times 10 = 9.8$	$32600 \times 10 = 326000$
GARDEN TRIMMING	10	60	6500	$\frac{100-60}{100} \times 10 = 4$	$6500 \times 10 = 65000$
TIMBER	05	30	18600	$\frac{100-30}{100} \times 5 = 3.5$	$18600 \times 5 = 93000$
TINCANS	05	03	700	$\frac{100-3}{100} \times 5 = 4.85$	$700 \times 5 = 3500$
				78.45	1474000

$$\text{Unit energy for 100 kg of sample} = \frac{1474000}{100} = \underline{14740 \text{ kJ/kg}}$$

$$\begin{aligned} \text{Energy content for dry basis kJ/kg} &= 14740 \times \frac{100}{100 - \% \text{ moisture content}} \\ &= 14740 \times \frac{100}{100 - 21.55} \\ &= \underline{18789.0376 \text{ kJ/kg}} \end{aligned}$$

$$\begin{aligned} \% \text{ moisture content} &= \frac{a-b}{a} \times 100 \\ &= \frac{100 - 78.45}{100} \times 100 \\ &= \underline{21.55} \end{aligned}$$

$$\begin{aligned} \text{Energy content for ash} &= 14740 \times \frac{100}{100 - \% \text{ moisture content} - \% \text{ ash}} \\ &= 14740 \times \frac{100}{100 - 21.55 - 5} \\ &= \underline{20068.073 \text{ kJ/kg}} \end{aligned}$$

- 2 Estimate energy content of solid waste with following composition also estimate total energy content on dry basis and ash free dry basis. Take sample as 1000 kg and ash % as 7%.

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COMPONENT	% BY MASS	MOISTURE CONTENT	ENERGY CONTENT (KJ/Kg)	DRY MASS (Kg)	TOTAL ENERGY CONTENT
Food waste	45	70	4650	13.5	209250
paper	05	06	16750	4.7	83750
card board	15	05	16300	1.425	244500
plastics	15	02	32600	14.7	489000
wood	20	20	18600	16	372000

$\Sigma = 63.15$ ~~34200~~
1398500

$$\text{Unit energy} = \frac{1398500}{100} = 13985 \text{ KJ/kg}$$

$$\% \text{ moisture content} = \frac{1000 - 63.15}{100} \times 100 = 936.85$$

$$\begin{aligned} \text{Energy content for dry basis KJ/kg} &= 13985 \times \frac{1000}{1000 - 936.85} \\ &= 1547.33 \text{ KJ/kg} \end{aligned}$$

$$\begin{aligned} \text{Energy content for ash basis KJ/kg} &= 13985 \times \frac{1000}{1000 - 936.85 - 7} \\ &= 1555.07 \text{ KJ/kg} \end{aligned}$$

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